Anatomy

The Digestive System

The **alimentary canal** consists of the buccal cavity, pharynx, oesophagus, duodenum, ileum and the rectum which leads to the cloaca and opens outside by the cloacal aperture. The wide mouth opens into the buccal cavity. On the floor of the **buccal cavity** lies a large **muscular sticky tongue**. The tongue is attached in front and free behind. The free edge is forked. When the frog sights an insect it flicks out its tongue and the insect gets glued to the sticky tongue. The tongue is immediately withdrawn and the mouth closes. A row of small and pointed **maxillary teeth** is found on the inner region of the upper jaw (Figure. 4.17) In addition **vomerine teeth** are also present as two groups, one on each side of the internal nostrils. The lower jaw is devoid of teeth. The mouth opens into the buccal cavity that leads to the **oesophagus** through the **pharynx**. Oesophagus is a short tube that opens into the stomach and continues as the intestine, rectum and finally opens outside by the cloaca (Figure 4. 18). **Liver** secretes bile which is stored in the gall bladder. **Pancreas**, a digestive gland produces pancreatic juice containing digestive enzymes.

Food is captured by the bifid tongue. Digestion of food takes place by the action of **Hydrochloric acid** and **gastric juices** secreted from the walls of the stomach. Partially digested food called chyme is passed from the stomach to the first part of the intestine, the duodenum. The duodenum receives bile from the gall bladder and pancreatic juices from the pancreas through a common bile duct. **Bile** emulsifies fat and **pancreatic juices** digest carbohydrates, proteins and lipids. Final digestion takes place in the intestine. Digested food is absorbed by the numerous finger-like folds in the inner wall of **intestine** called **villi** and **microvilli.** The undigested solid waste moves into the **rectum** and passes out through the **cloaca.**

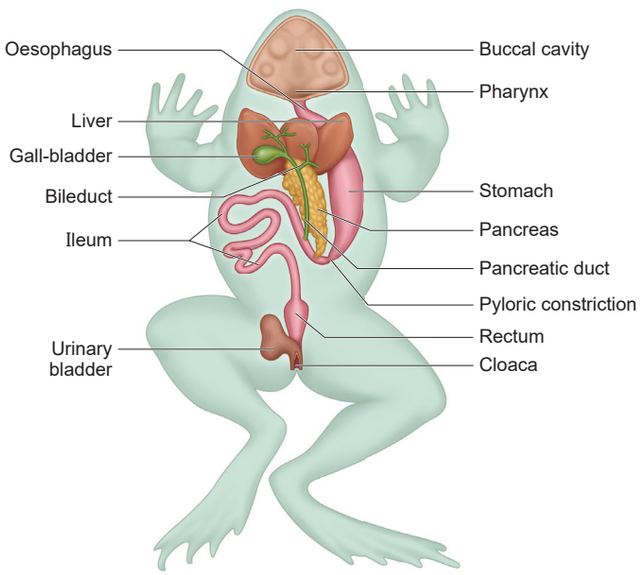


Figure: 4.18 Digestive System of *Rana hexadactyla*



Anus: The opening at the lower end of the alimentary canal in mammals through which solid waste leaves the body.

Cloaca: The common chamber into which the intestinal, urinary and genital tracts open. It is present in birds, reptiles, amphibians, elasmobranch fishes and monotremes. The cloaca has an opening for expelling its contents from the body and in females it serves as the depository for sperm.

Respiratory System

Frog respires on land and in the water by two different methods. In water, **skin** acts

TN_GOVT_XI_Zoology_Ch04.indd 65 09-01-2020 5.14.26 PM